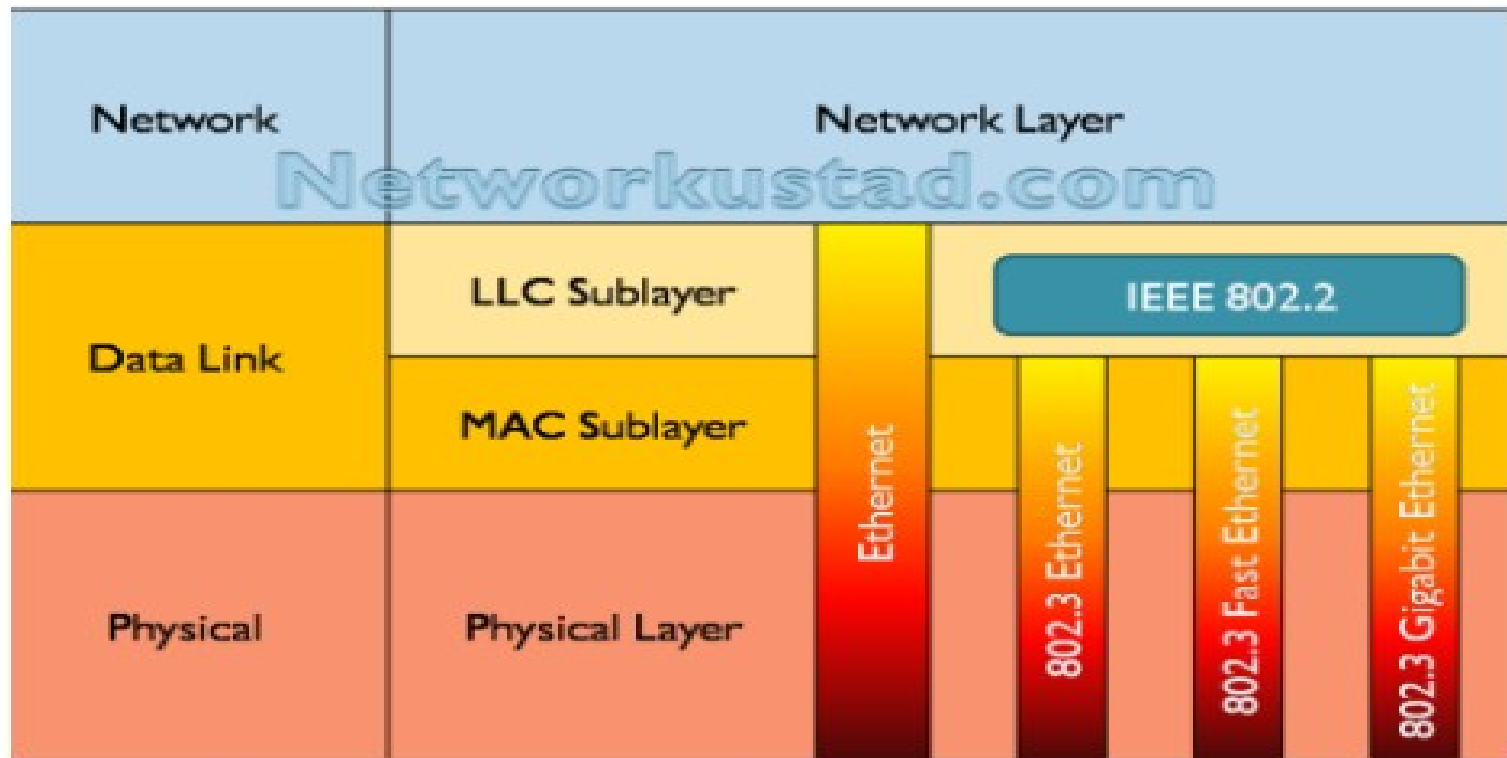


Chapter 3-part two

Medium Access Control

Data link Sub layer



- We can consider the data link layer as two sub layers. The upper sub layer is responsible for data link control, and the lower sub layer is responsible for resolving access to the shared media.

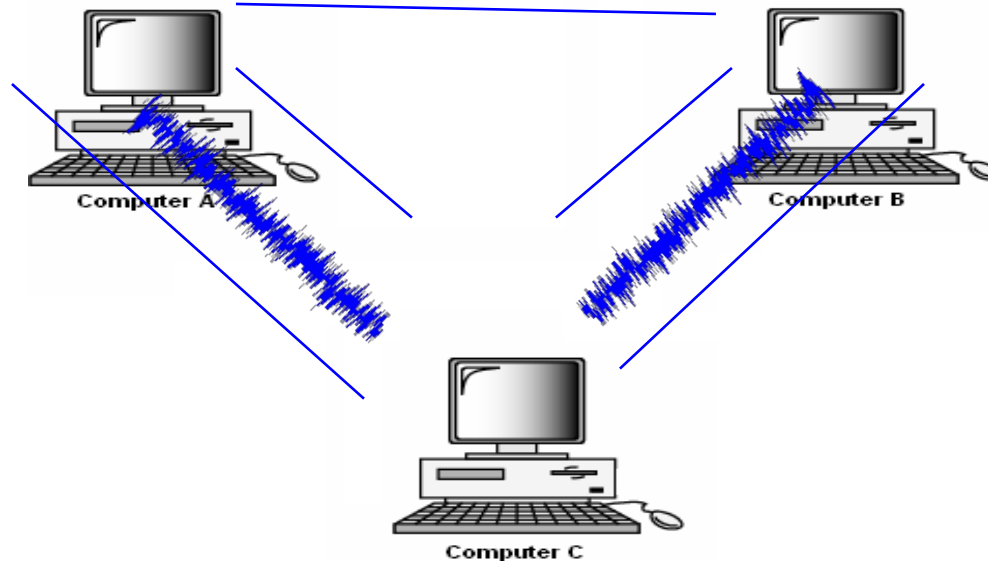
- Link Types
 - Point-to-Point(PPP)
 - Broadcast

Point-to-point networks

Point-to-point networks are those in which, when a message is sent from one computer to another, it usually has to be sent via other computers in the network.

PPP for dial-up access

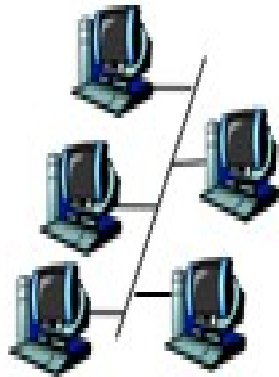
point-to-point link between Ethernet switch and host



Broadcast (shared wire or medium)

old-fashioned Ethernet

802.11 wireless LAN



shared wire (e.g.,
cabled Ethernet)



shared RF
(e.g., 802.11 WiFi)



shared RF
(satellite)



humans at a
cocktail party
(shared air, acoustical)

Broadcast networks

Broadcast networks have a single communication channel that is shared by all the machines on the network.



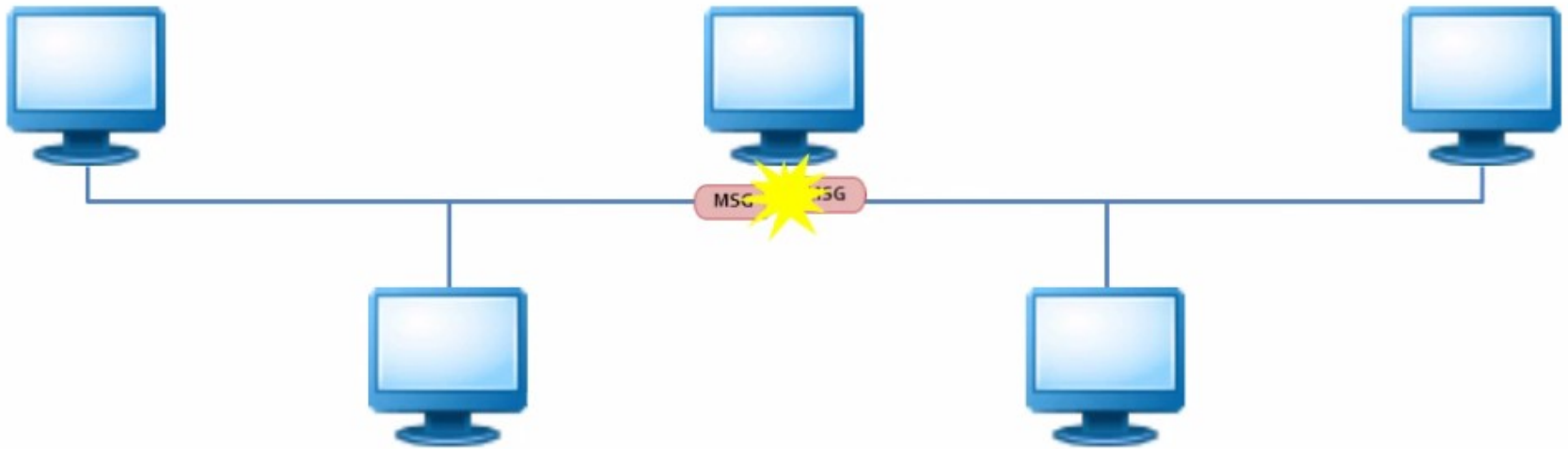
Shared channel

Why Multiple Access Protocol ?

- If there is a dedicated link between the sender and the receiver then data link control layer is sufficient. However if there is **no** dedicated link present then multiple stations can access the channel simultaneously. This leads to **collision**.
- Hence multiple access protocols are required to decrease **collision** and avoid **crosstalk**.

Medium Access Control

Broadcast Network



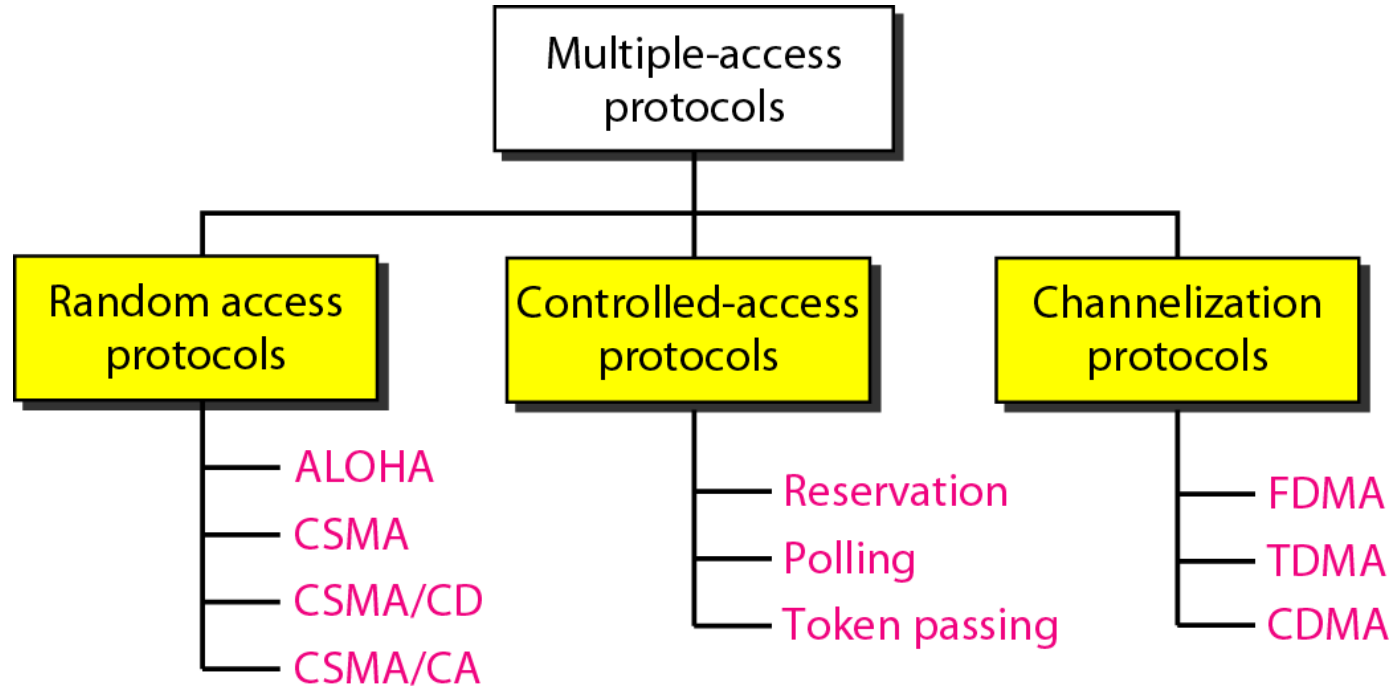
if two or more devices are sending data at the same time without checking the status of the channel whether the channel is busy or idle the data will collide with each other and become unusable

Multiple Access protocols

- distributed algorithm that determines how nodes share channel, i.e., determine when node can transmit
- communication about channel sharing must use channel itself!
 - no out-of-band channel for coordination

- If more than one station attempts to send at the same time there is an access conflict/collision and frames are either destroyed/lost/or modified.

Types of Multiple –access protocol



Random Access

- All stations have same superiority that is no station has more priority than another station.
 - **No station is superior.**
 - Each station has the right to the medium without being controlled by any other station. No station assigned control over another
 - No station does permit/not permit another station to send.
 - The decision is based on the state of the medium . Any station can send data depending on medium's state(**idle or busy**)

- Every station transmits whenever it feels like it.
- There is no scheduled time for a station to transmit.
- Stations compete (contention) on who takes the medium next.
- There is no agreement on who transmits next.
If more than one station tries to send at the same time there is an access conflict (collision) and that the frames will be either destroyed or modified.

- To avoid access conflict , we need to follow a procedure.
- Any procedure for multiple access must answer the following questions
 - WHO is going to use the channel ?
 - WHEN can the station access the medium?
 - WHAT can the station do if the medium is busy?
 - HOW can the station determine the success for failure of the transmission?
 - WHAT can the station do if there is an access conflict?
 - For HOW much time the channel is used ?
- Examples: ALOHA, CSMA, CSMA/CD

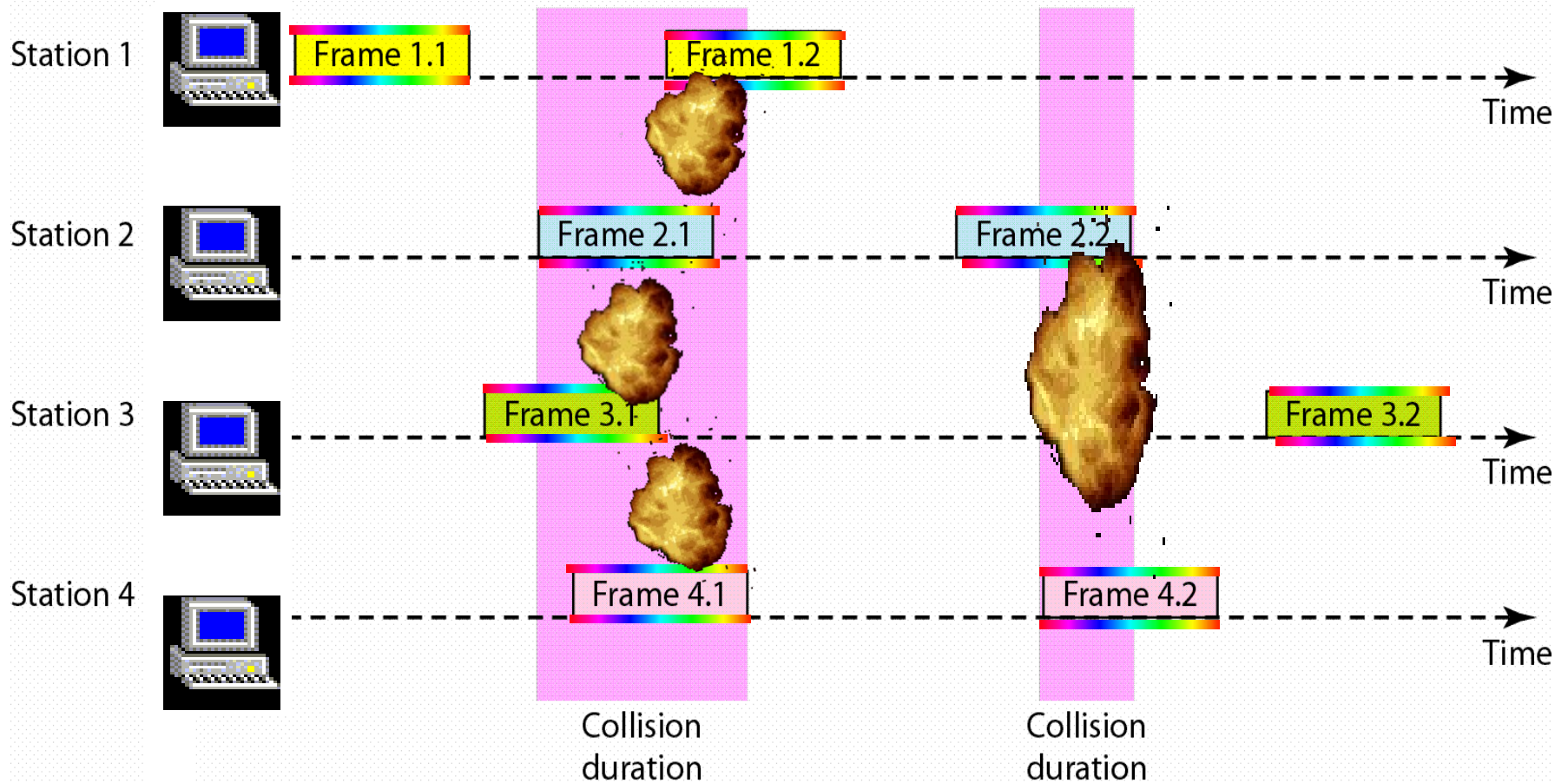
ALOHA

- was developed at the University of Hawaii in early 1970.
- Is a random access protocol
- Designed for a wireless LAN but it is also applicable for shared medium.
- In this, multiple stations can transmit data at the same time and can hence lead to collision and data being garbled.
- There are two versions of Aloha system which differ with respect to whether or not time is divided up into discrete slots into which all frames must fit.
 - PURE ALOHA
 - SLOTTED ALOHA

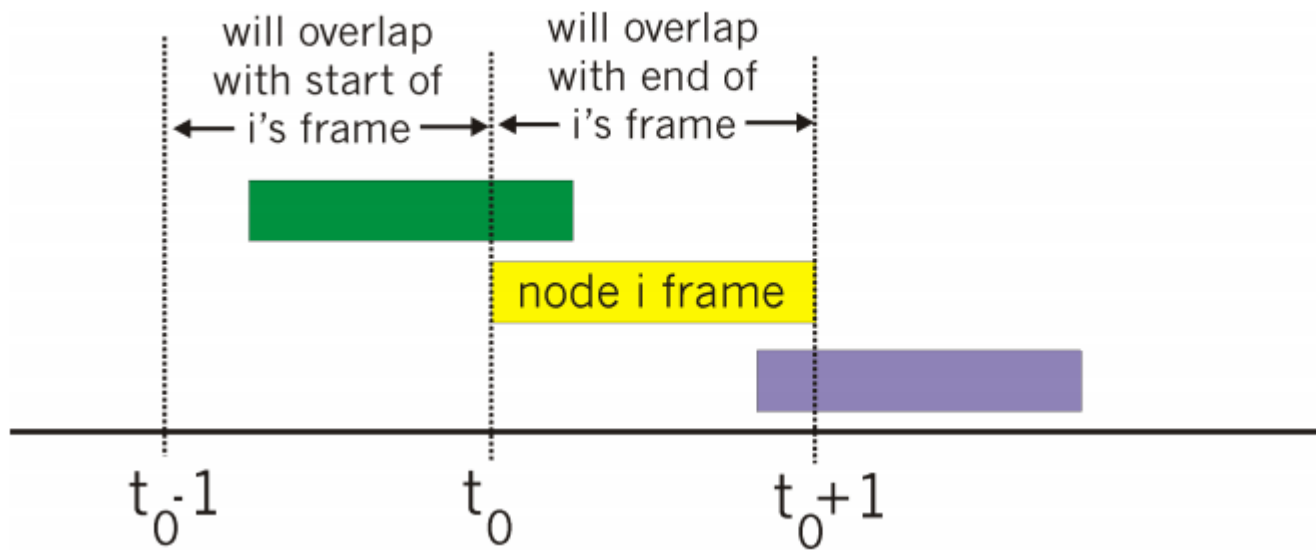
Pure (unslotted) ALOHA

- Each station sends a frame whenever it has a frame to send.
- Since there is only one channel, there is a
- possibility of collisions.
- simpler, no synchronization

Pure Aloha



- Pure ALOHA allows the stations to transmit whenever they have the data to be sent.
- Pure ALOHA did not have a restriction on when stations can send. This means that stations could start spamming, sending just after/before another station has finished sending stuff.
- Whenever two frames try to occupy the channel at the same time, there will be a **collision** and both will be garbled.
- If the first bit of the frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will have to be retransmitted later.



collision probability: frame sent at t_0 collides with other frames sent in $[t_0 - 1, t_0]$

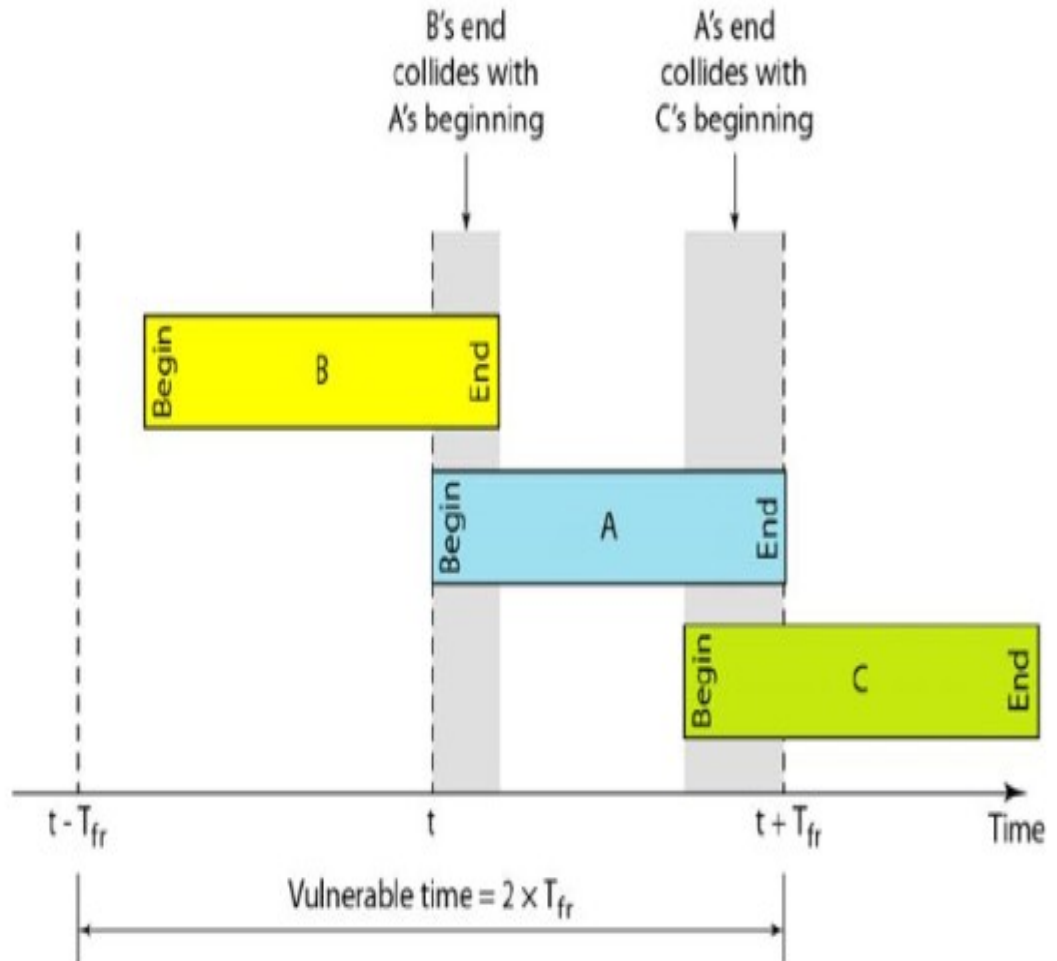
Vulnerable time for ALOHA protocol

Vulnerable Time: The length of time in which there is a possibility of collision.

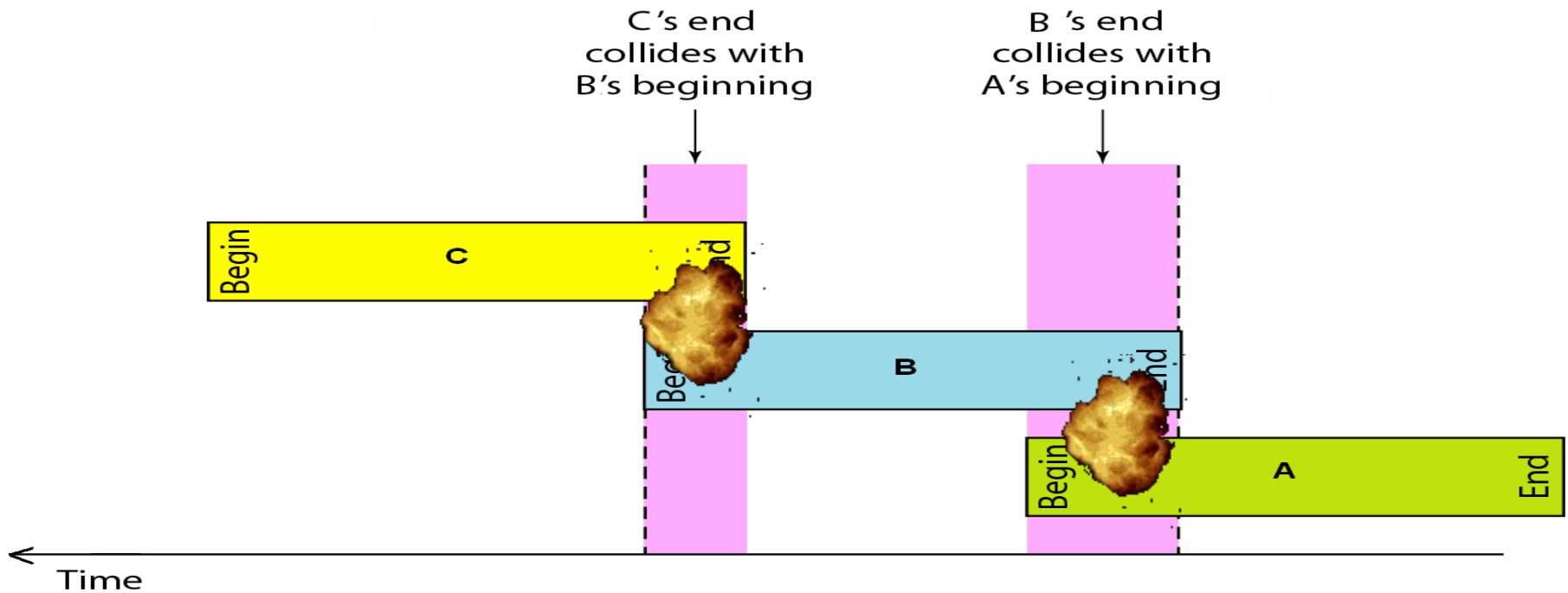
Station A sends a frame at time t . Imagine Station B has sent a frame between $t - T_{fr}$ and t .

This leads to a collision between frames from Stations A and B. The end of B's frame collides with the start of A's frame.

Suppose C sends a frame between $t + T_{fr}$ and t . Here a collision occurs between the end of A's frame and the start of C's frame.



frame transmission time (T_{fr})



If the first bit of a new frame overlaps with just the last bit of a frame almost finished, both the frames will be totally destroyed. It does not distinguish between a total loss or a near miss.

- When a station sends data it waits for an ACK.
- If the ACK does not arrive in a specific time period, then the sender just re-sends the frame.
- **Problem:** If two stations re-send frames at the exact same times on the same medium, both frames will be lost. (Hello Race Condition!)
- **Solution:** Pure ALOHA dictates that a sender, upon failing to get an ACK (ACK timed out), will wait a random amount of time before re-sending the frame. This is the back-off time, T_B

- Since different stations wait for different amount of time, the probability of further collision decreases.
- What if the channel is really crappy and it gets filled with re-sends? This creates congestion.
 - Pure ALOHA prevents congesting the channel with re-transmitted frames.
 - After a maximum number of re-transmission attempts ($K_{\max}$), a station must give up and try again later.
- The throughput of pure aloha is maximized when frames are of uniform length.

Procedure for Pure ALOHA Protocol

Constraints/Figures:

- **Time-out Period:**
 - = maximum possible round-trip delay.
- **Back-off Time T_B**
 - Typically related to K (maximum re-transmission attempts).
 - Implementation Specific.

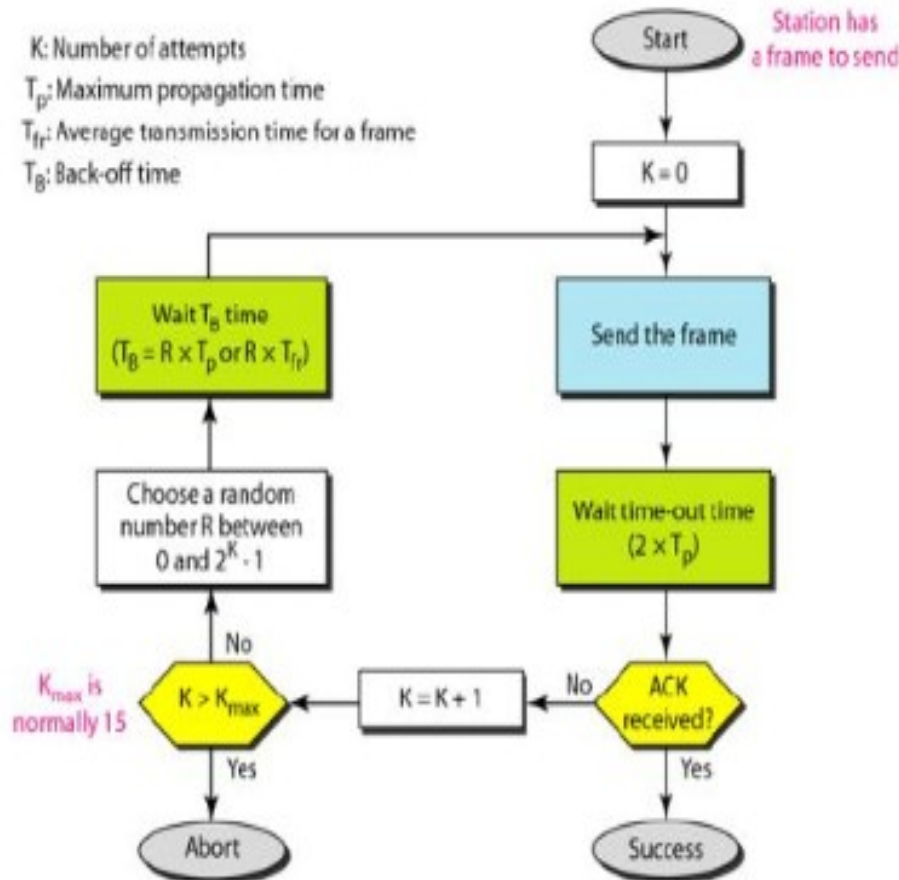
Back-off Time Calculation.

In this course, we have a formula for calculating T_B

Procedure:

1. Randomly pick a number between $0 \dots 2^K - 1$. Call this number Z. (Where Z is a discrete range of integers).
2. $T_B = Z * T_p$, where T_p is the maximum propagation time.
3. OR $T_B = Z * T_{rr}$, where T_{rr} is the average time required to send out a frame.

The value of K_{MAX} is typically chosen as 15.



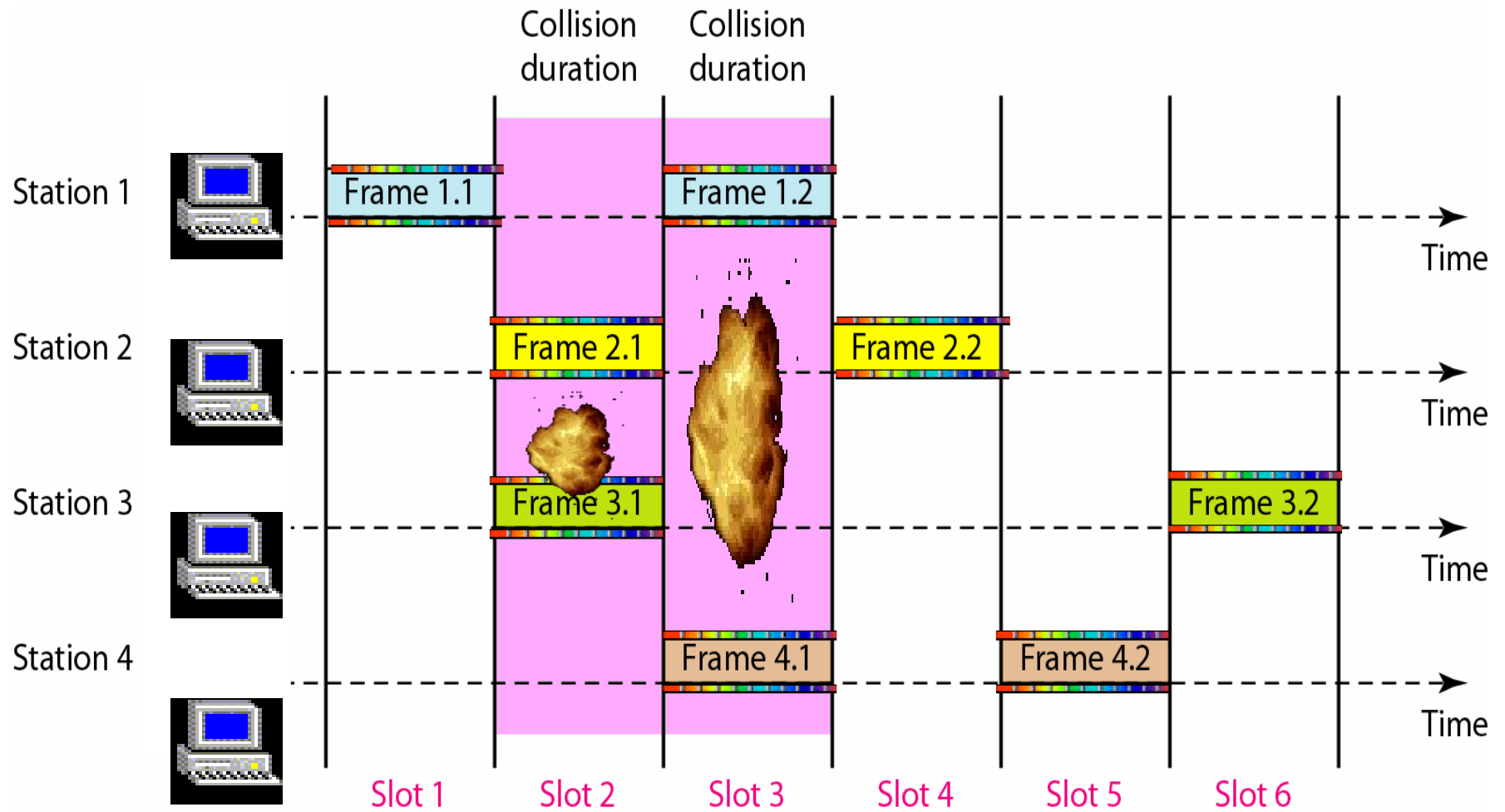
- **Vulnerable Time**= $2 * T_{tr}$
- **Throughput**= $G * e^{-2g}$;where G is the number of stations wish to transmit at the same time.
- **The maximum throughput**=0.184 for $G=0.5(1/2)$

Slotted ALOHA

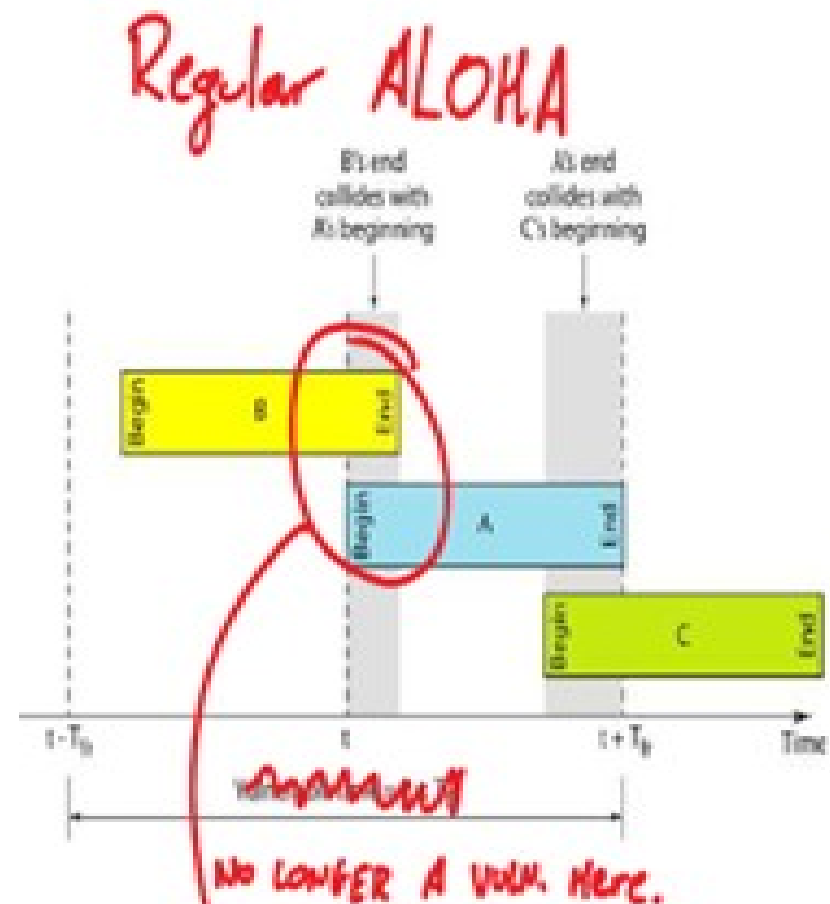
- It was developed to just improve the efficiency of pure aloha as the chances for collision in pure aloha are high.
- The time of the shared channel is divided in to discrete time intervals called **slots** of size frame transmission time (T_{fr}), each corresponding to one packet.
- These timeslots are agreed upon by all stations.
- all frames same size
- nodes are synchronized
- sending of data is allowed **only at the beginning** of the timeslot.

Slotted ALOHA

- if 2 or more nodes transmit in same slot, all nodes detect collision operation:
- when node obtains fresh frame, transmits in next slot
- if no collision: node can send new frame in next slot
- if collision: node retransmits frame in each subsequent slot with prob= p until success

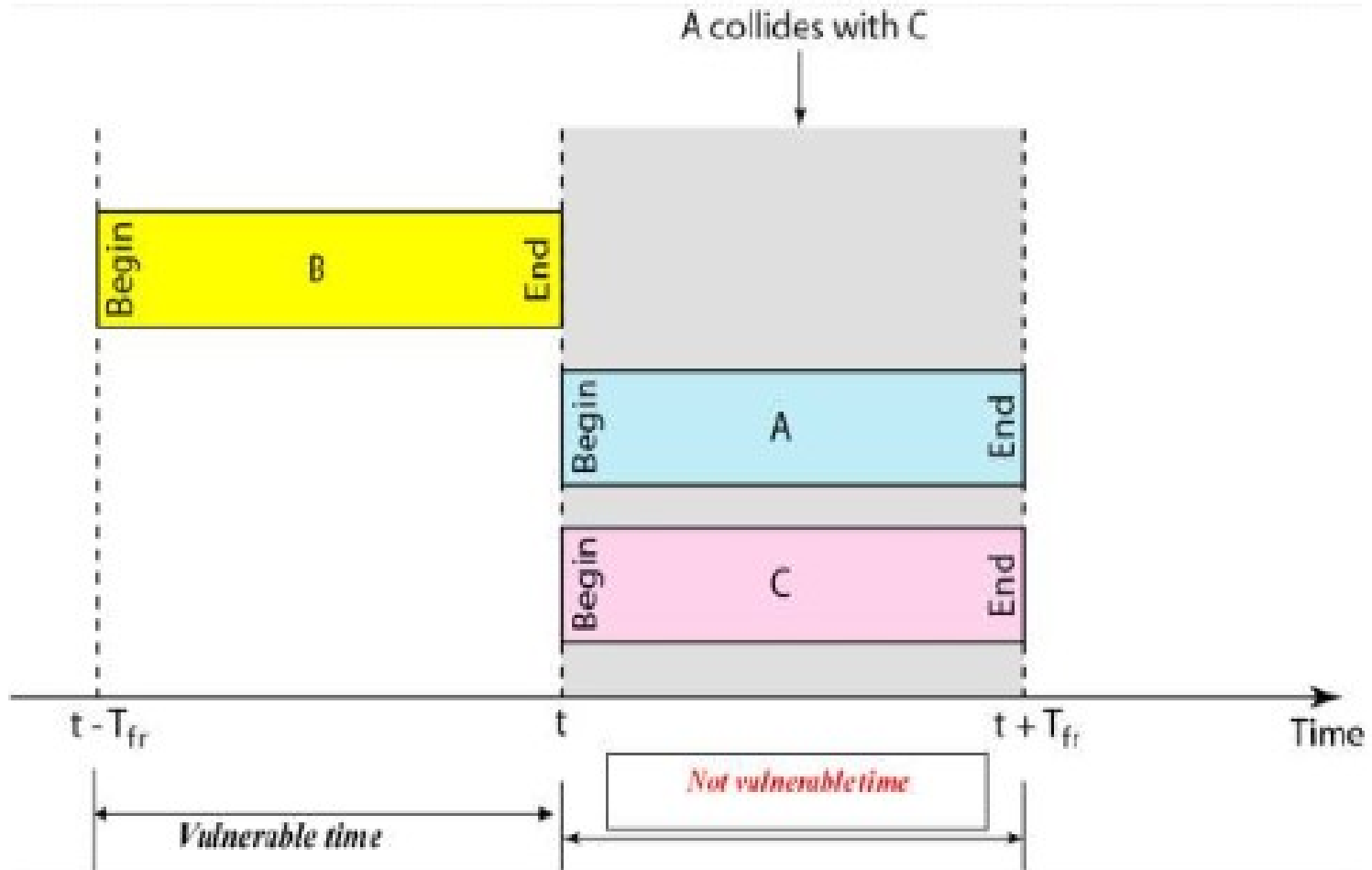


- A station is allowed to send at the beginning of a time slot. If it misses the start of the slot, it must wait until the beginning of the next time slot. This reduces the probability of collision.
- This implies the station which started at the beginning of the time slot has already finished sending the frame.
 - This means that we no longer have a vulnerability as shown



Slotted ALOHA Vulnerable Time = T_{PK}

Vulnerable time for slotted ALOHA protocol



- **Vulnerable Time**= T_{tr}
- **Throughput**= $G * e^{-G}$;where G is the number of stations wish to transmit at the same time.
- **The maximum throughput**=0.368 for $G=1$

Pros:

- single active node can continuously transmit at full rate of channel
- highly decentralized: only slots in nodes need to be in sync
- simple

Cons:

- collisions, wasting slots
- idle slots
- nodes may be able to detect collision in less than time to transmit packet
- clock synchronization

Differences between Pure ALOHA and slotted ALOHA

Pure ALOHA	Slotted ALOHA
Any Station can transmit data at any time.	Any station can transmit the data only at the beginning of any time slot.
Vulnerability time = 2 x Transmission time.	Vulnerability time = Transmission time.
The probability of successful transmission in Pure ALOHA is $G \times e^{-2G}$	The probability of successful transmission in Slotted ALOHA is $G \times e^{-G}$
The max probability of successful transmission in pure aloha occurs at $G=\frac{1}{2}$ and the value is $\frac{1}{2} \times e^{-1} = 18.39\%$	The max probability of successful transmission in slotted aloha occurs at $G=1$, the value is $1 \times e^{-1} = 36.79\%$
Time is continuous and not synchronised.	Time is discrete and Synchronised.

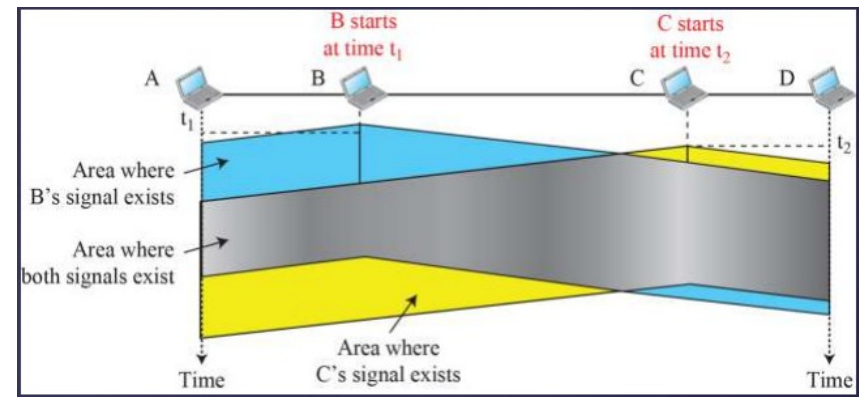
Carrier Sense Multiple Access(CSMA)

- The chance of collision can be reduced if a station senses the medium(Carrier) before trying to use it.
- Principle of CSMA:
 - Each station first listen to the medium (aka: check the state) before sending
 - **"Sense before transmit"** or **"Listen before talk"**
 - Carrier busy=Transmission is taking place
 - Carrier idle=no transmission currently taking place.
- CSMA can reduce possibility of collision, but not eliminate it.

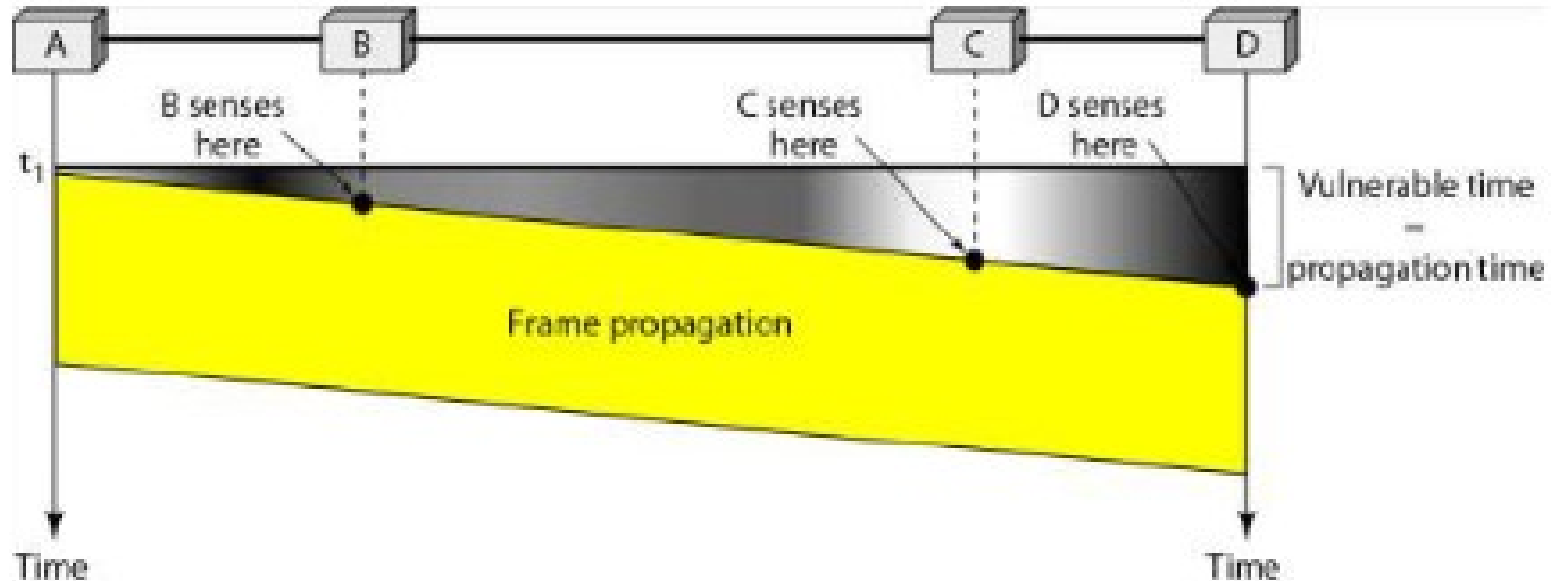
- The possibility of collision still exists because of **propagation delay**. When a station sends a frame, it still takes time for the first bit to reach every station and for every station to sense it.
- A station may sense the medium and find it idle, only because the first bit sent by another station had not been received (due to **propagation delay**)

- Types of CSMA
 - 1-Persistent CSMA
 - P-Persistent CSMA
 - Non-Persistent CSMA
 - o-Persistent CSMA
- The modified version
 - CSMA/CD(CSMA with Collision Detection)
 - CSMA/Ca(CSMA with Collision Avoidance)

- at time T_1 , station B senses the medium and finds it idle, so it sends a frame.
- At time $T_2 > T_1$, station C senses the medium and finds it idle, because at time T_2 , the first bits from station B haven't made it to station C.
- Both stations have sent a frame at the "same-enough" time. Both frames are destroyed.



Vulnerable Time in CSMA



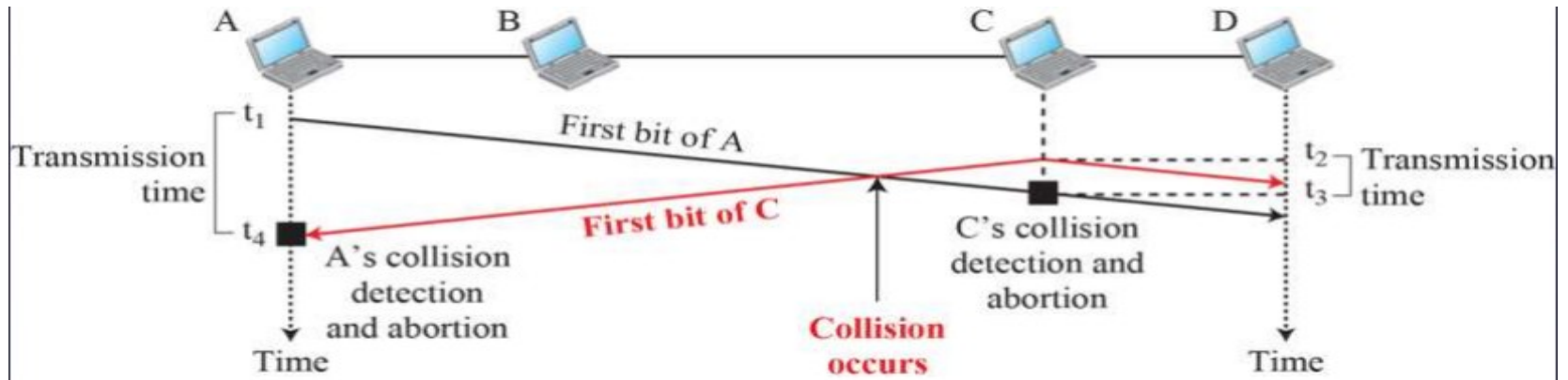
Carrier sense multiple access with collision detection(CSMA/CD)

- If two stations sense the channel to be idle and begin transmitting simultaneously, they will detect the collision almost immediately.
- Rather than finish transmitting their frames, which are irretrievably garbled anyway, they should abruptly stop transmitting as soon as the collision is detected.
- Quickly terminating damaged frames saves time and bandwidth.
- Is widely used on LANs in the MAC sub layer.
- Access method used by Ethernet: CSMA/CD

- At time point marked t_0 , a station has finished transmitting its frame.
- Any other station having a frame to send may now attempt to do so. If two or more stations decide to transmit simultaneously, there will be a collision.
- Collision can be detected by looking at the power of pulse width of the received signal and comparing it to the transmitted signal.

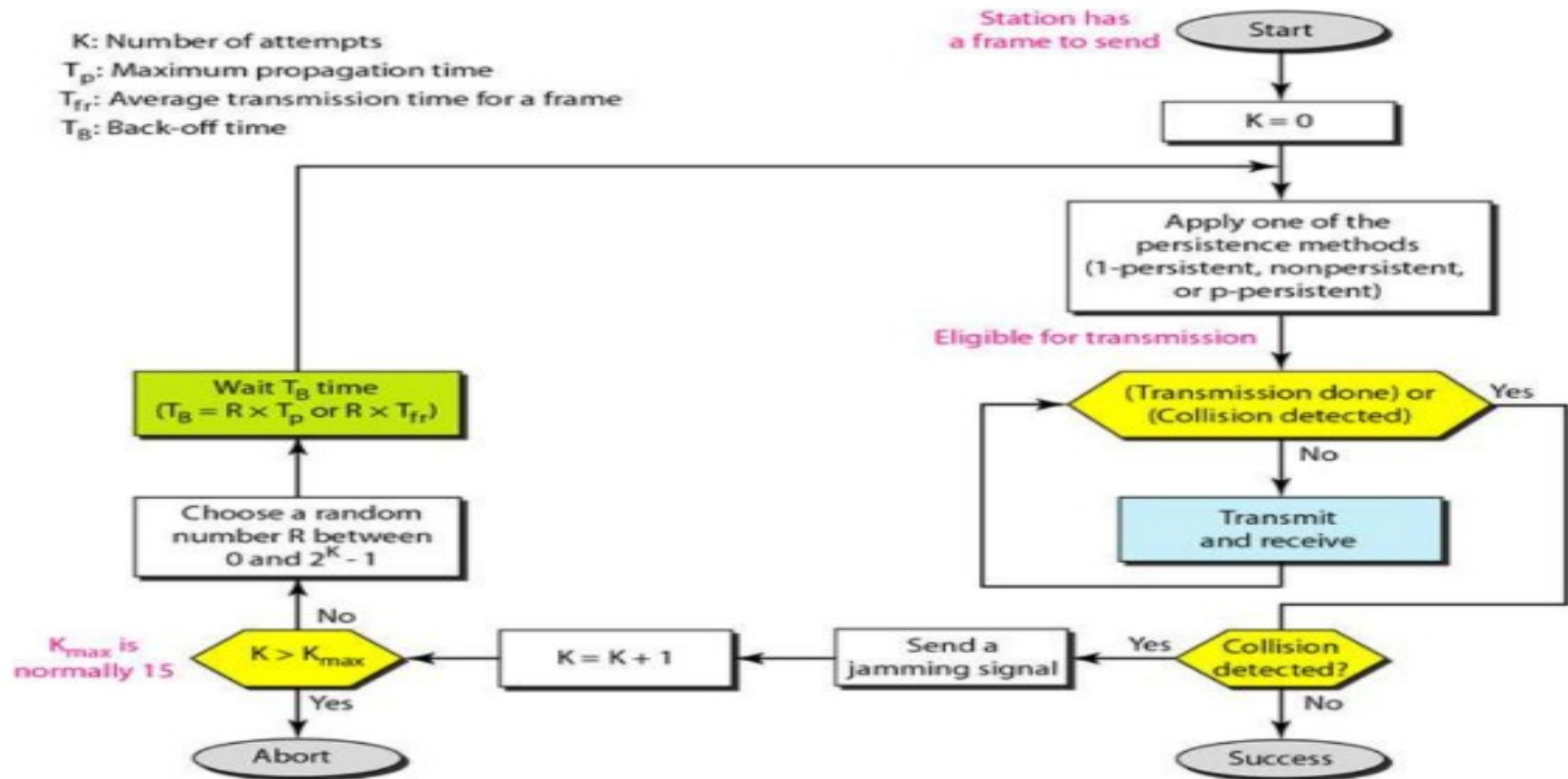
- After the station detects a collision, it aborts its transmission, waits a random period of time, and then tries again, assuming that no other station has started transmitting in the meantime.
- Therefore, model for CSMA/CD will consist of alternating contention and transmission periods, with idle periods occurring when all stations are quiet.

Collision of the first bits in CSMA/CD



1. at time t_1 A starts sending bits of the frame
2. At time t_2 , C hasn't sensed the first bits from A.
3. C starts sending bits, which go left and right.
4. Sometime after t_2 , a collision occurs.
5. At time t_3 , C detects a collision when it received part of A's frame.
 - a. (Sending things.... Oh shit! We have some of A's frame before mine should get anywhere!)
6. C immediately stops transmission.
7. A detects collision at time t_4 when it gets part of C's frame. A immediately aborts transmission.
8. We see that
 - a. A was spamming for time $t_4 - t_1$
 - b. A.C was spamming for time $t_3 - t_2$

Flow Diagram for CSMA/CD



ALOHA: first send entire frame, then wait for ACK

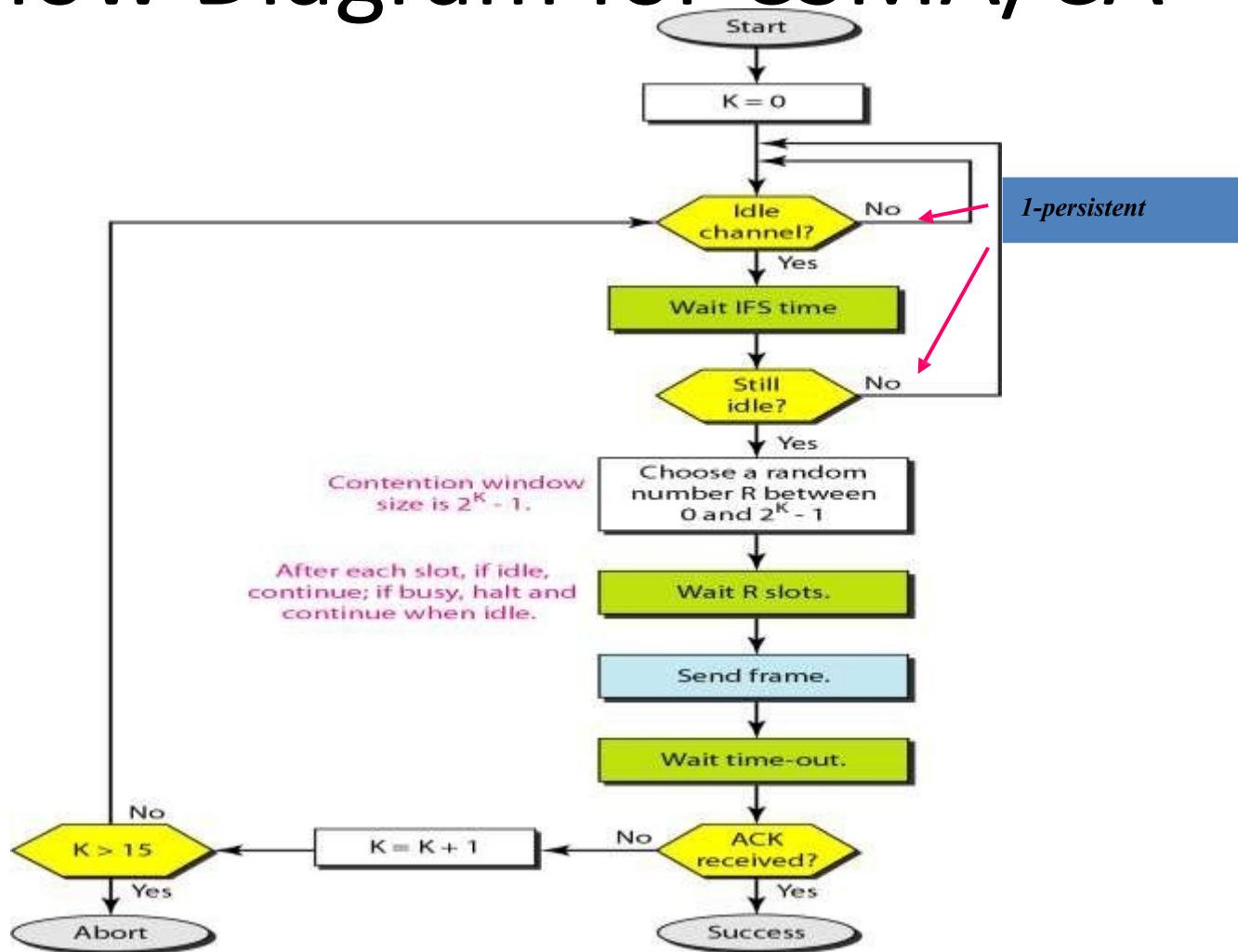
CSMA/CD: send and detect simultaneously. No ACK is needed.

Carrier sense multiple access with collision avoidance(CSMA/CA)

- Is a multiple access method in which carrier sensing is used, but nodes attempting to avoid collisions by beginning transmission only after the channel is sensed to be “**idle**”.
- It is particularly important for wireless networks, where the collision detection for the alternative CSMA/CD is not possible due to wireless transmission desensing their receivers during packet transmission.
- In a wireless network, the received signal has very little energy due to energy loss in transmission a collision may add only 5~10% additional energy not useful for effective collision detection

- CSMA/CA is unreliable due to the hidden node problem and exposed terminal problem.
- Solution: RTS/CTS exchange Features
- Operates at the data link layer (layer 2) of the OSI model
- used in 802.11 wi-fi

Flow Diagram for CSMA/CA



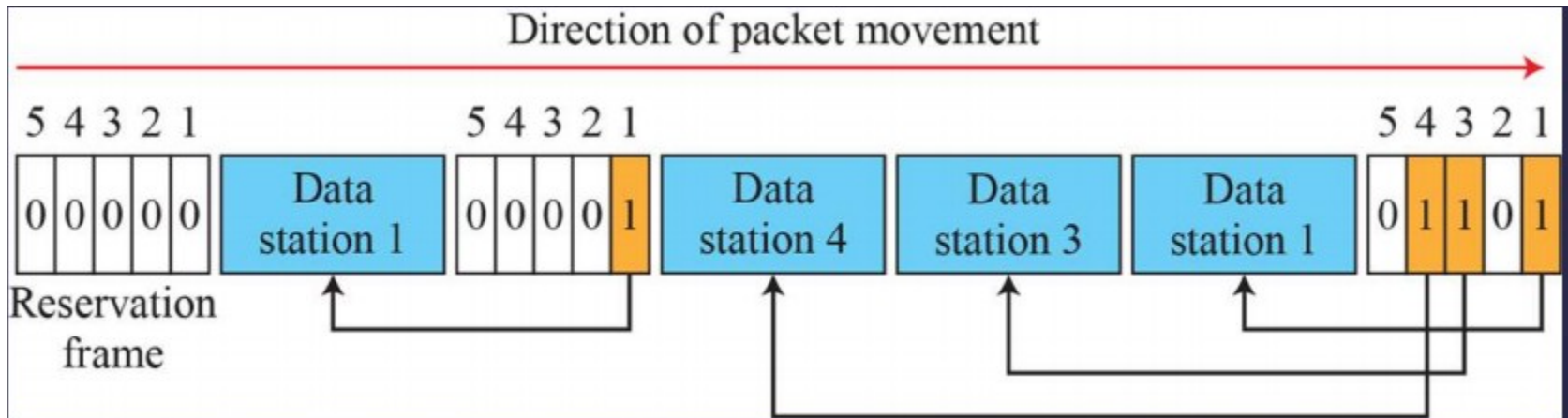
Controlled-Access Protocols

- In controlled access, the stations consult on another to find which station has the right to send.
- A station cannot send unless it has been authorized by other stations.
- Controlled access methods
 - **Reservation**
 - Ex. DQDB, Cable modem, ...
 - **Polling**
 - Select and Poll
 - Ex. HDLC (normal response mode)
 - **Token passing**
 - Ex. Token ring (FDDI), token bus, ...

Reservation

- A station need to make a reservation before sending data.
- Time is divided into intervals
- In each interval, a reservation frame precedes the data frames sent in that interval.
- If there are N stations in the system, there are exactly N reservation mini slots in the reservation frame
- Each minislot belongs to a station

- When a station needs to send a data frame it makes a reservation in its own minislot
- Stations that have made reservation can send their data frames.



- In the diagram above, 5 stations, 5 minislot reservation frame
- At first interval,
 - Only stations 1,3,4 have made reservation.
- In second interval,
 - only station 1 has made a reservation

Pooling

- Requires one of the nodes to be designated as a primary station (Master node) , and the others are secondary stations.
- The master node polls each of the nodes in a round-robin fashion.
- All data exchanges must be made through the primary device even when the ultimate destination is a secondary device.
 - The primary device controls the link, the secondary devices follow its instructions.
 - It is up to the primary device to determine which device is allowed to use the channel at a given time.
- The primary device, therefore, is always the initiator of a session

- The master node first sends a message to node 1, saying that it can transmit up to some maximum number of frames.
- After node 1 transmits some frames, the master node 2 can transmits some frames to the maximum number of frames.
- The master node can determine when the nodes has finished sending its frame by observing the lack of a signal on the channel.

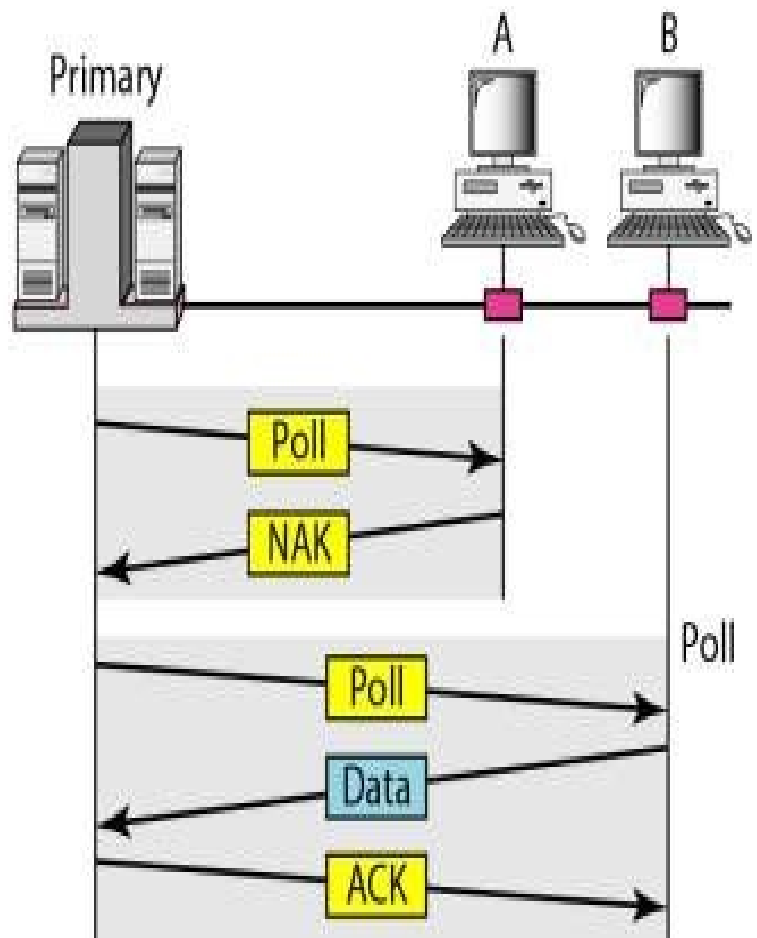
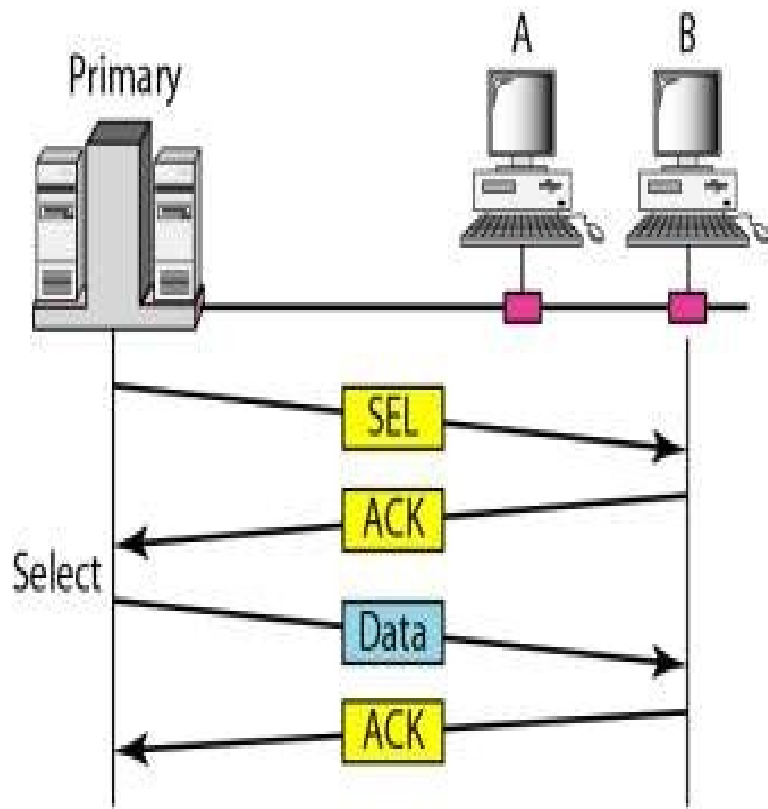
- The procedure continues in the manner, with the master node polling each of the nodes in a cyclic manner.
- Eliminates collision
- This allows polling to achieve a much higher efficiency

drawback

- The protocol introduces a pooling delay-the amount of time required to notify a node that it can transmit.
- If the master node fails, the transmit entire channel becomes inoperative.

Pooling –Functions

- If the primary wants to receive data, ask the secondary's if they have anything to send **poll function**
- If the primary wants to send data, tells the secondary's to get ready to receive --> **select function**.
- Select
 - The primary alerts a secondary that data is coming.
 - Needs an ACK from the secondary so that the receiver is ready. (The ACK is in response to a SEL frame).
 - SEL is like "YOU'RE GONNA GET DATA WHEN YOU'RE READY. YOU READY BRO?"
- POOL
 - Primary asks everyone in turn if they have something to send. If NAK, then ask the next station



Efficiency of pooling

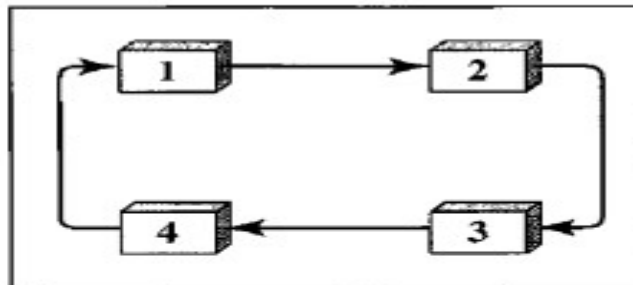
- Let T_{poll} be the time for polling and T_t be the time required for transmission of data. Then
- $\text{Efficiency} = T_i / (T_t + T_{\text{poll}})$

Token Passing

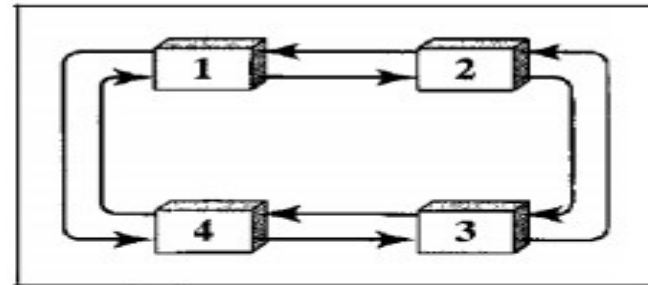
- A station is authorized to send data when it receives a special frame called a token
- Here there is no master node
- A small, special-purpose frame known as a token is exchanged among the nodes in some fixed order
- Stations are arranged around a ring (physically or logically)
 - A token circulates around the ring
- When a node receives a token, it holds onto the token only if it has some frames to transmit; otherwise, it immediately forwards the token to the next node.

- If the node have frames to transmit
 - it waits for the token.
 - The station captures the token and sends one or more frames as long as the allocated time has not expired.
 - It releases the token to be used by the successor station.

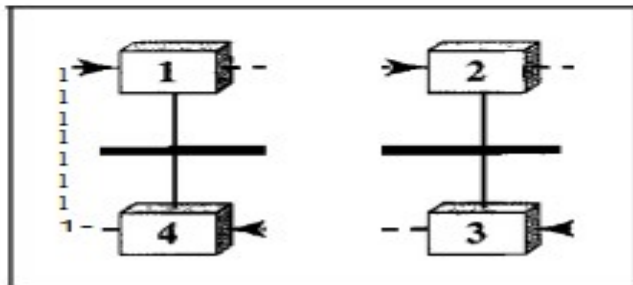
Logical ring and physical topology in token-passing access method



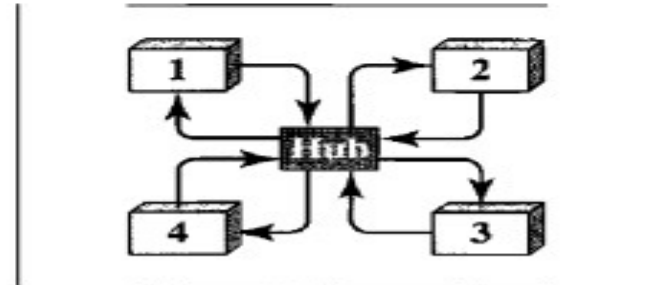
a. Physical ring



b. Dual ring

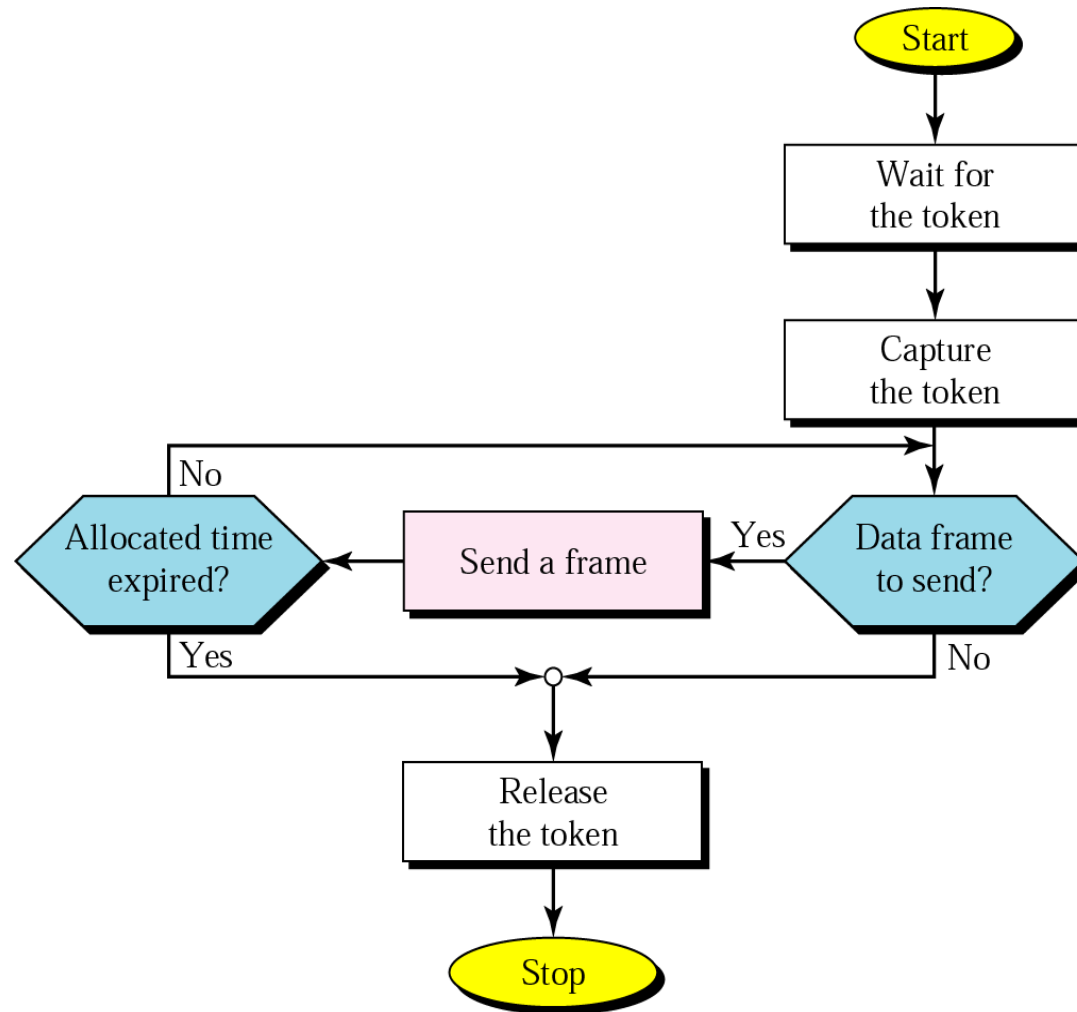


c. Bus ring



d. Star ring

- Procedure



- Token passing is decentralized and highly efficient. but it has problems as well.
- For example, the failure of one node can crash the entire channel. Or if node accidentally neglects to release the token, then some recovery procedure must be invoked to get the token back in circulation

Fixed Assignment /Channelization

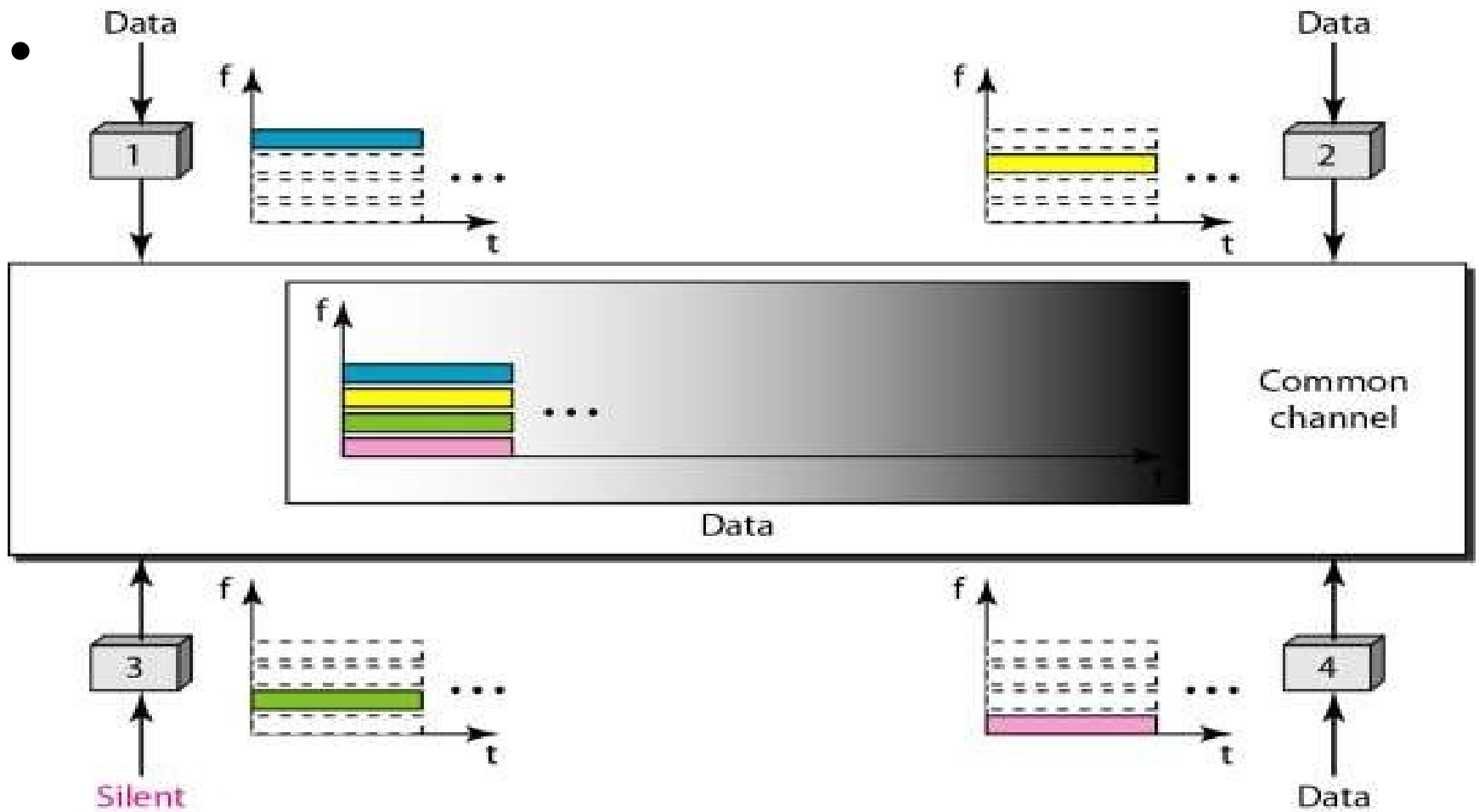
- A Multiple-access method in which the available BW of a link is shared in time, frequency, or through code, between different stations.
- Allocate the channel in a static manner. This allocation scheme doesn't depend on the demands of the nodes

Multiplexing

- Multiplexing in a computer networking means multiple signals are combined together thus travel simultaneously in a shared medium.
- Multiplexing= sharing the bandwidth
- Types
 - Frequency Division Multiplexing(FDMA)
 - Time Division Multiplexing(TDMA)
 - Code division Multiplexing (CDMA)
- **Features**
 - FDMA, TDMA are based on FDM and TDM in the physical layer
 - CDMA is a data link multiple-access protocol

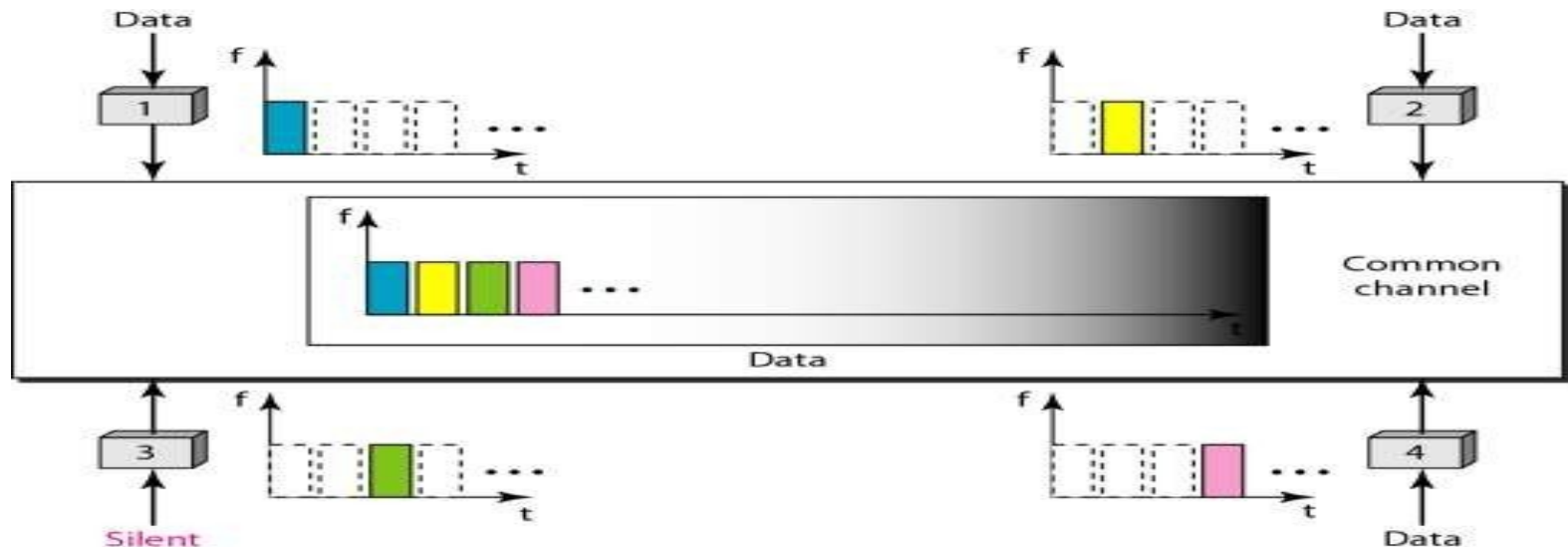
Frequency-Division Multiple Access (FDMA)

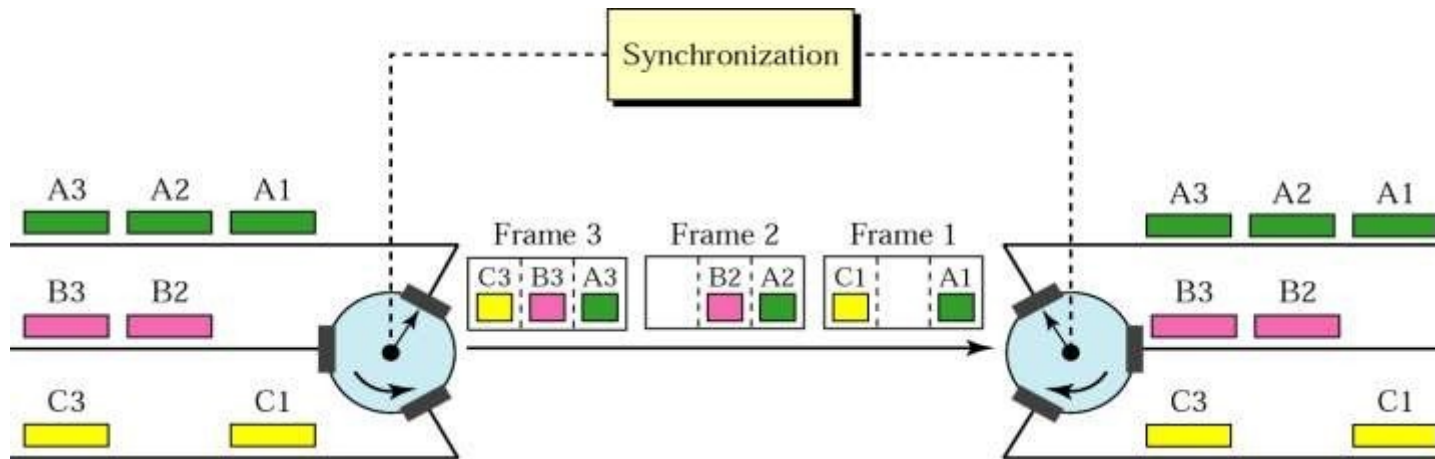
- A data link layer protocol that uses FDM at the physical layer
- The available BW of the common channel is divided into bands that are separated by guard bands.
- The available bandwidth is shared by all stations.
 - Each stations uses its allocated band to send its data
 - To prevent station interferences, the allocated bands are separated from one another by small *guard bands*.
Ex. AMPS (1G cellular networks)



Time-Division Multiple Access (TDMA)

- A data link layer protocol that uses TDM at the physical layer
- The stations share the capacity of the channel in time
- The entire bandwidth is just one channel
- Each station is allocated a time slot during which it can send data
 - Ex. GSM (2G cellular networks - Europe)



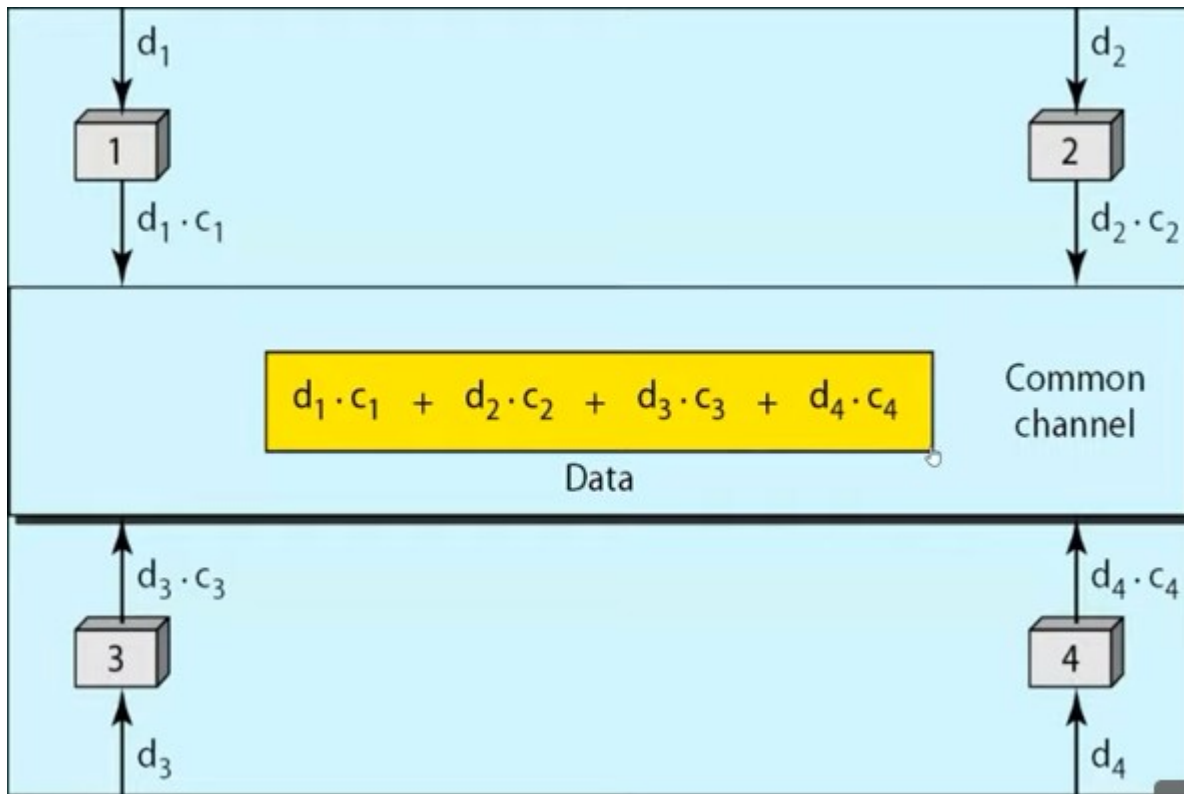


The main problem with TDMA lies in achieving synchronization between the different stations. Each station needs to know the beginning of its slot and the location of its slot.

This may be difficult because of propagation delays introduced in the system if the stations are spread over a large area. To compensate for the delays, we can insert *guard times*. Synchronization is normally accomplished by having some synchronization bits (normally referred to as preamble bits) at the beginning of each slot.

Code Division Multiple Access (CDMA)

- In CDMA, one channel carries all the transmissions simultaneously.
- It differs from FDMA because only one channel occupies the entire bandwidth of the link
- It differs from FDMA because all the stations can send data simultaneously there is no time sharing.
- used in several wireless broadcast channels (cellular, satellite, etc) standards
- unique “code” assigned to each user; i.e., code set partitioning



- The assigned code have two properties
 - If we multiply each code by another we get 0
 - If we multiply each code by itself, we get 4 (the number of stations)
 - Example
 - $\text{Data} = (d_1c_1 + d_2c_2 + d_3c_3 + d_4c_4$